Arnol Manuel Fokam

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Research Engineer at InstaDeep, London. I build reinforcement-learning environments, train agents inside them, and build the tooling that shows what those agents actually did. Our routing agents kept failing a subset of printed circuit boards. I tried to solve those boards by hand, established they were unsolvable, wrote detectors for the features that made them so, and swept the 15,000-board training set: 24 were contaminated. The environments were at fault, not the agent. I also replaced my research team's per-developer evaluation branches with one pipeline that scores any checkpoint on 30+ boards it has never seen, and built a rollout viewer that replays the observation and the action distribution at each timestep.

Skills

- **Reinforcement learning:** multi-agent reinforcement learning, training environments, synthetic environment generation, quality-diversity search
- **LLM and agent systems:** multi-agent orchestration, tool calling, retrieval, sandboxed code execution
- **Evaluation and tooling:** evaluation pipelines, benchmark harnesses, trajectory visualisation, automated failure detection, training-data auditing
- **Languages and tools:** Python, PyTorch, JAX, NumPy, Docker, Git, Linux

Experience

Research Engineer, InstaDeep (a BioNTech company), London, UK | Apr 2026 to Present

- Traced routing failures on DeepPCB to a synthetic generator producing unsolvable boards, by hand-solving the cases the agent never cleared, then wrote detectors sweeping the 15,000-board training set. 24 were contaminated: the environments were at fault, not the agent.
- Consolidated my research team's per-developer evaluation branches into one zero-shot pipeline that scores any model checkpoint on 30+ boards it has never seen.
- Built a rollout visualisation tool showing what an agent observed and the action distribution it produced at each timestep, so a failure can be read off the trajectory rather than from a score; the approach prompted further visualisers on my research team.
- Train multi-agent reinforcement-learning agents that place components and route printed circuit boards of up to 8 layers and 1,200 connections, reducing an hours-long manual design task to minutes.

Research Engineer, InstaDeep (a BioNTech company), Kigali, Rwanda | Aug 2024 to Mar 2026

- Supervised a project benchmarking five DeepSeek-R1 reasoning models, from 1.5B to 32B, against OpenAI's o-series on MLE-bench, building the evaluation harness and serving the open models with vLLM.
- Built a three-agent system that proposes, implements and interprets experiments on the cluster, unattended. It ran 50 experiments in two days, where a researcher would implement two or three.
- Built the log-streaming engine behind Laila, demonstrated at InstaDeep's AI Day 2024, turning hundreds of messages a minute from four lab instruments into equipment state queryable in plain English by tool-calling.
- Contributed to RL research published as a **NeurIPS 2025 Oral**.

Engineering Intern, InstaDeep (DeepPack team), Remote, South Africa | Aug 2023 to Mar 2024

- Built a synthetic environment generator for the reinforcement-learning agent behind DeepPack, InstaDeep's bin-packing software, using a quality-diversity algorithm so the generated packing problems stayed varied instead of collapsing onto one type.

Projects

- **Bibby**: helped build the early version of an AI-first LaTeX editor that drafts equations from plain language, finds and formats citations, and reviews drafts; now used by over 5,900 researchers.

Publications

- **Breaking the Performance Ceiling in Reinforcement Learning requires Inference Strategies**. Felix Chalumeau, Daniel Rajaonarivonivelomanantsoa, Ruan de Kock, Claude Formanek, Sasha Abramowitz, Oumayma Mahjoub, Wiem Khlifi, Simon Du Toit, Louay Ben Nessir, Refiloe Shabe, **Arnol Fokam**, Siddarth Singh, Ulrich Mbou Sob, Arnu Pretorius. *NeurIPS 2025 (Oral)*.
- **How Temporal Correlations Shape Memory in Linear Recurrent Neural Networks**. **Arnol Manuel Fokam**, Fasseu Sieyondji Akpevwoghene, Edem Fiifi Dawson. *AAAI 2027, under review. Independent research, carried out outside of work*.
- **Oryx: a Scalable Sequence Model for Many-Agent Coordination in Offline MARL**. Claude Formanek, Omayma Mahjoub, Louay Ben Nessir, Sasha Abramowitz, Ruan de Kock, Wiem Khlifi, Daniel Rajaonarivonivelomanantsoa, Simon Du Toit, **Arnol Fokam**, Siddarth Singh, Ulrich Mbou Sob, Felix Chalumeau, Arnu Pretorius. *NeurIPS 2025*.

- **Hierarchically Composing Level Generators for the Creation of Complex Structures**. Michael Beukman, **Manuel Fokam**, Marcel Kruger, Guy Axelrod, Muhammad Nasir, Branden Ingram, Benjamin Rosman, Steven James. *IEEE Transactions on Games, 2023*.
- **Analysing Cross-Lingual Transfer in Low-Resourced African Named Entity Recognition**. Michael Beukman, **Manuel Fokam**. *IJCNLP-AACL 2023*.

Education

University of the Witwatersrand, Johannesburg, South Africa | Feb 2022 – Sep 2024
MSc Computer Science
Thesis: A Novel Data Augmentation Framework for Contrastive Self-Supervised Learning.

University of the Witwatersrand, Johannesburg, South Africa | Feb 2021 – Dec 2021
BSc (Hons) Computer Science, *cum laude*
Thesis: Influence of Contrastive Learning on Source Code Plagiarism Detection.

University of Buea, Buea, Cameroon | Oct 2016 – Aug 2019
BSc Applied Physics (Minor in Computer Science)
Thesis: Introduction to Quantum Computing (Theory and Applications).

Awards and Service

- **DeepMind Scholarship**, Wits, 2022–2023 · *AI by you* video
- **Runner-up**, Minecraft Open-Endedness Challenge, GECCO, 2022
- **Deep Learning Indaba**: ML practicals tutor (Rwanda, 2025); RL workshop organiser (Tunisia, 2022)
- **Teaching Assistant** (Reinforcement Learning, NLP), Wits, 2022–2023
- **Research mentorship**: led two early-career collaborators to an AAAI 2027 submission